



Homework 1



(Advanced) Data Mining: Algorithms and Applications-Winter 2024

Due on Feb 7, 11.59PM



Important

- Please create a GitHub repository to submit your work, push/submit needed files to GitHub, and provide a link on Canvas.
- Please put all images and their explanations in a single pdf file and submit it along with an R script which has all R commands that you used.
- I recommend working on a R notebook which you can output to a pdf file by using Rstudio (Knit option), but you don't have to.
- Please leave comments in your R script
- Please do not email your work.



Datasets for this homework can be found at the link below and feel free to add this folder to your Google Drive:

<https://drive.google.com/drive/u/0/folders/1ehWunuAo7CE1Vk2JYkUnQMmxh5pph3C>

1. Use "Su_raw_matrix.txt" for the following questions (30 points).
 - (a) Use `read.delim` function to read `Su_raw_matrix.txt` into a variable called `su`. (Notice that `su` has become a data frame now)
 - (b) Use `mean` and `sd` functions to find mean and standard deviation of `Liver_2.CEL` column.
 - (c) Use `colMeans` and `colSums` functions to get the average and total values of each column.
2. Use `rnorm(n, mean = 0, sd = 1)` function in R to generate 10000 numbers for the following (`mean`, `sigma`) pairs and plot histogram for each, meaning you need to change the function parameter accordingly. Then comment on how these histograms are different from each other and state the reason. (20 points)
 - (a) `mean=0, sigma=0.2`
 - (b) `mean=0, sigma=0.5`

Please save your figures as image from RStudio. (Hint: to see the difference in plots you may need to set the `xlim` parameter in plot function to `c(-5,5)`)

3. Perform the steps below with "dat" dataframe which is just a sample data for you to observe how each plot function (3b through 3e) works. Notice that you need to have `ggplot2` library installed on your system. Please refer slides how to install and import a library. Installation is done only once, but you need to import the library every time you need it by saying `library(ggplot2)`. Then run the following commands for questions from 3a through 3e and observe how the plots are generated first. (20 points)
 - (a)

```
dat <- data.frame(cond = factor(rep(c("A","B"), each=200)),
  rating = c(rnorm(200),rnorm(200, mean=.8)))
```
 - (b)

```
# Overlaid histograms
ggplot(dat, aes(x=rating, fill=cond)) +
  geom_histogram(binwidth=.5, alpha=.5, position="identity")
```
 - (c)

```
# Interleaved histograms
ggplot(dat, aes(x=rating, fill=cond)) + geom_histogram(binwidth=.5, position="dodge")
```
 - (d)

```
# Density plots
ggplot(dat, aes(x=rating, colour=cond)) + geom_density()
```
 - (e)

```
# Density plots with semitransparent fill
ggplot(dat, aes(x=rating, fill=cond)) + geom_density(alpha=.3)
```
 - (f) Read "diabetes_train.csv" into a variable called `diabetes` and apply the same functions 3b through 3e for the `mass` attribute of `diabetes` and save the images. (Hint: instead of `cond` above, use the `class` attribute to color your groups. When you have fill option, your plots should show same type of chart for both groups in different colors on the same figure. Keep in mind that `diabetes` and `dat` are both DataFrames)
4. Read the `titanic.csv` file from DATA folder to a variable named `passengers` and perform the following steps and explain the operation very briefly. Please make sure you have tidyverse installed on your system and you may specifically need to import the `tidyr` library. Otherwise, the chain of operations through "piping" won't work. (20 points):
 - (a) `passengers %>% drop_na() %>% summary()`
 - (b) `passengers %>% filter(Sex == "male")`

- (c) `passengers %>% arrange(desc(Fare))`
 - (d) `passengers %>% mutate(FamSize = Parch + SibSp)`
 - (e) `passengers %>% group_by(Sex) %>% summarise(meanFare = mean(Fare), numSurv = sum(Survived))`
5. By using `quantile()`, calculate 10th, 30th, 50th, 60th percentiles of skin attribute of diabetes data. (10 points)